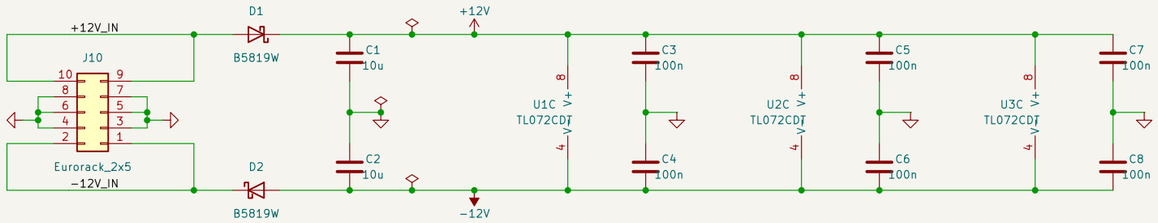
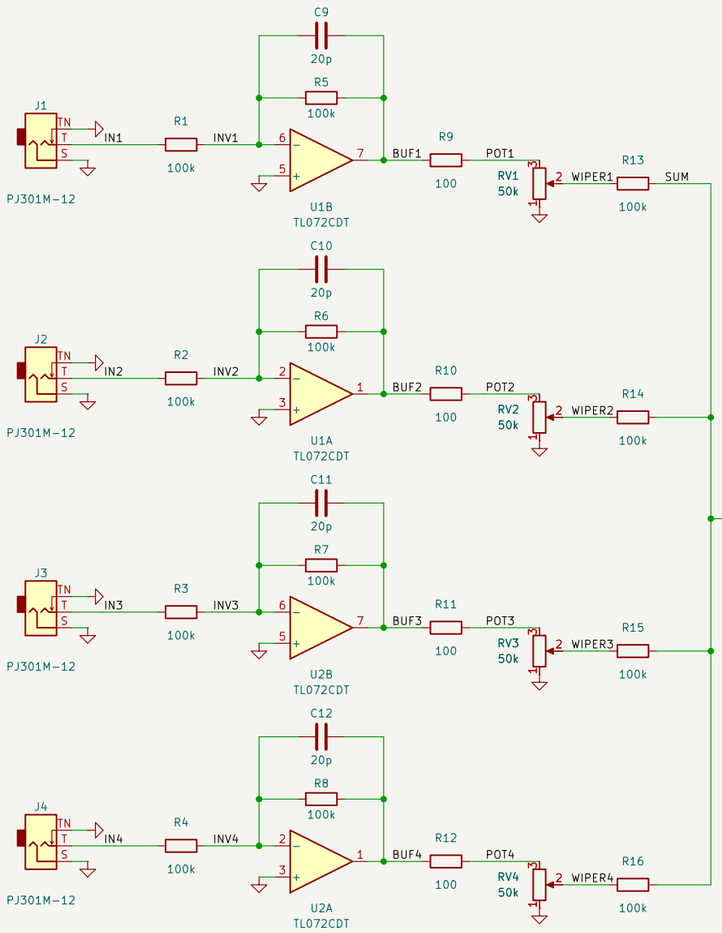
BIG MODULAR IMIX — S3 schematic: 4-channel DC-coupled buffered mixer (CV or audio). Circuit intent: docs/research.md

POWER: Eurorack +12V / −12V -> B5819W per rail -> +12V / −12V (about +/−11.6 V). J10 +5V unused.
C3..C8: 100n at every op-amp supply pin; C1/C2: 10u bulk.

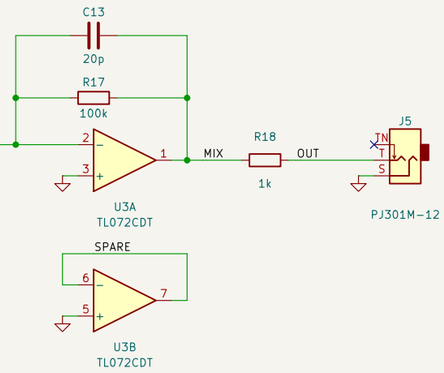


PWR_FLAG: #FLG01 marks the Eurorack GND; #FLG02/#FLG03 mark the +12V/−12V rails,
fed from the header only through D1/D2, which ERC cannot see through.

CHANNELS 1-4 (Bread Modular imix topology): jack tip -> 100k -> inverting buffer (100k || 20p, gain −1)
-> 100R -> RV pin 3 (CW); pin 1 (CCW) = GND; wiper -> 100k -> SUM. Input impedance 100k at any knob
position; the buffer drives the pot from ~0 ohm. TN grounds an unpatched input. Full CW = unity gain.



MIXER: U3A sums (−1 x each wiper) back to the input polarity; 1k series out.
OUT = WIPER1 + WIPER2 + WIPER3 + WIPER4, clipping near +/−10 V (TL072 on +/−11.6 V).
U3B unused: grounded follower.



Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S3 — 4-channel buffered DC mixer, +/−12 V, 6HP — not released, not supplier-accepted

Big Modular

Sheet: /
File: imix.kicad_sch

Title: Big Modular — IMIX

Size: A3	Date:	**Rev: S3**
KiCad E.D.A. 10.0.6		Id: 1/1